INCH-POUND
MIL-DTL-60945D
21 November 2006
SUPERSEDING
MIL-DTL-0060945C (AR)
05 April, 2004
MIL-C-60945B (AR)
12 May, 1993

DETAIL SPECIFICATION

CARTRIDGE, 40MM - PROJECTILE ASSEMBLY, PRACTICE, M385A1

This Document Dated 21 November 2006 is Active For New Design

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Cartridge, 40MM - Projectile Assembly, Practice, M385A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806 5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1310

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8886326 - Cartridge, 40MM: Practice, M385A1
8866714 - Projectile Assembly, For Cartridge, 40MM: Practice, M385A1
8866715 - Body

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

(Copies of this document are available online at http://www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 M385 projectile assembly and components.

3.2.1 Body. The body including the salt spray test shall be in accordance with drawing 8866715.

3.2.2 Projectile assembly. The projectile assembly including the push test of rotating

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band test shall be in accordance with drawing 8866714.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix

Method of verification
N/A - Not Applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 – Test

Classes of verification
A - First Article
B - Conformance

SECTION 3 REQUIREMENT	DESCRIPTION	VERIFICATION METHODS					VERIFICATION CLASS		SECTION 4 VERIFICATION METHOD
		N/A	1	2	3	4	A	B	
3.1	FIRST ARTICLE			X	X	X	X		4.3
3.2	PROJECTILE ASSEM & COMPONENTS			X	X	X	X	X	4.3, 4.4
3.2.1	BODY			X	X	X	X	X	4.4.2.1, 4.5.1
3.2.2	PROJECTILE ASSEMBLY			X	X	X	X	X	4.4.2.2, 4.5.2
3.3	AMMUNITION LOT NUMBERING				X		X	X	4.5.4
3.4	WORKMANSHIP				X		X	X	4.5.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection Condition. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First Article. When specified, a sample of the Projectile Assembly, 40MM: Practice, M385A1 shall be subjected to first article inspection in accordance with Table II and Table III.

4.3.1 First Article Quantity. The first article sample shall consist of the following items in sample quantities as indicated in Table II.

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TABLE II. First article quantity

PART DESCRIPTION	DRAWINGS	QUANTITY
Body	8866715	50
Projectile Assembly	8866714	50

4.3.2 Inspections to be performed. The first article inspection shall be 100 % examination of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed and the tests specified in Table III herein.

TABLE III. First article inspection

	EXAMINATION OR TEST	N0. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD
	Body (Dwg. 8866715)			
	Salt Spray	10	3.2.1	4.5.1
	Projectile Assembly (Dwg. 8866714)			
	Push test of rotating band	50	3.2.2	4.5.2
Notes:				

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of projectiles shall be produced with components complying with the following constraints:

a. Metal parts or components from one interfix number, from one manufacturer

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as majors and Inspection Level II for all minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 and 4.4.2.2. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.4.2.1	TITLE Body	SHEET 1 OF 1	DRAWING NUMBER 8866715
			NEXT HIGHER ASSEMBLY 8866714

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Diameter of rotating band surface (after knurl)	Level IV	3.2.1	Gage
102.	Spherical radius	Level IV	3.2.1	Gage
103.	Location of basic diameter (excluding cut-off nib)	Level IV	3.2.1	Gage
104.	Run-out of indicated area on taper with body diameter	Level IV	3.2.1	Gage
105.	Body diameter	Level IV	3.2.1	Gage
106.	Diameter between knurl	Level IV	3.2.1	Gage
107.	Width of knurl crest	Level IV	3.2.1	Gage
108.	Width of rotating band surface	Level IV	3.2.1	Gage
109.	Length from back of rotating band surface to front end of nose	Level IV	3.2.1	Gage
110.	Location of bourrelet shoulder	Level IV	3.2.1	Gage
111.	Diameter of bourrelet	Level IV	3.2.1	Gage
112.	Diameter below bourrelet	Level IV	3.2.1	Gage
113.	Salt Spray 1/	2 per 8 hr shift	3.2.1	4.5.1
MINOR				
201.	Protective coating with bare spot on bottom or forward of rotating band groove	Level II	3.2.1	Visual
202.	Incorrect color	Level II	3.2.1	Visual
203.	Evidence of poor workmanship	Level II	3.4	Visual / 4.5.5

NOTES:
1/ The production represented by the 8-hour shift sample shall be rejected if any white corrosion is visible to the unaided eye at normal reading distance in any location on the band surface of the body, but not including the edges, when salt spray tested in accordance with 4.5.1.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.4.2.2	TITLE Projectile Assembly	SHEET 1 OF 1		DRAWING NUMBER 8866714
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Profile of rotating band	Level IV	3.2.2	Gage
102.	Diameter of rotating bands	Level IV	3.2.2	Gage
103.	Diameter of large groove	Level IV	3.2.2	Gage
104.	Diameter of small groove	Level IV	3.2.2	Gage
105.	Distance from nose to back of rotating band	Level IV	3.2.2	Gage
106.	Total run-out of rotating bands with bourrelet and body diameter	Level IV	3.2.2	Gage
107.	Total length (excluding cut-off nib)	Level IV	3.2.2	Gage
108.	Push test of rotating band	80 1/	3.2.2	4.5.2
MINOR				
201.	Marking missing	Level II	3.2.2	Visual
202.	Evidence of poor workmanship	Level II	3.4	Visual / 4.5.5

NOTES:
1/ Accept on zero (0) and reject on one (1) or more failures.

4.5 Methods of inspection.

4.5.1 Salt spray. The specified part shall be subjected to the salt spray test in accordance with ASTM B117. This test shall be conducted for 24 hours, or for the duration specified on the applicable drawing, using a salt solution of 5 percent. This test is a destructive test. Parts tested shall not be returned to the lot. Sampling and rejection criteria is as specified in the applicable referencing paragraph for each part.

4.5.2 Push test of rotating band on projectile assembly. The projectile assembly shall be placed in an approved fixture and the force specified on Dwg. 8866714 shall be applied. Observation shall be made for the requirement specified. This test is a destructive test. Parts tested shall not be returned to the lot. Deformation of the chamfered leading edge of the rotating band caused by compression in the test fixture is permissible. Movement as specified on Dwg. 8866714 is defined as movement of the back face of the rotating band relative to the body. Every time there is a change in the knurling configuration, a First Article Test shall be preformed. The body assemblies shall be subjected to temperature conditioning testing. For the ambient portion of the test, the bodies shall be conditioned at 70 $\pm$ 5 F at least eight (8) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the high temperature test, the body assemblies shall be conditioned at 125 $\pm$ 5 F at least four (4) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the low temperature test, the body assemblies shall be conditioned at - 25 $\pm$ 5 F at least four (4) hours, and

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then tested within fifteen (15) minutes after being removed from the temperature soak. Fifty (50) body assemblies shall be tested for each hot, cold and ambient portions of this test. Failure to meet the requirement on the specified drawing shall be the reason for rejection of the First Article. In addition the following ballistic testing are required to support government consideration of an alternative knurling configuration:

Rnds	Temperature	Test	Range	Inspection
144	ambient	single shot	200 feet	recover and inspect
50	hot	single shot	200 feet	recover and inspect
50	cold	single shot	200 feet	recover and inspect
96	ambient	rapid fire	200 feet	(5-6 round bursts) recover and inspect after each burst

- Single shot rounds must be recovered and inspected for a missing rotating band or movement of the rotating band and to ensure that the engraving does not cut into the aluminum knurling. Rounds should be recovered frequently to avoid the possibility of hitting each other.
- Rapid fire rounds must be recovered and inspected for movement of the rotating band and to ensure that the engraving does not cut into the aluminum knurling.. Rounds should be recovered after each burst.
- Additionally, any gun stoppage during the rapid fire must have the cause determined and recorded.
- Accept/reject criteria on the ballistic testing is accept on 0, reject on 1 for rotating band movement.

4.5.4 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

4.5.5 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements, examinations and tests for the Projectile Assembly for the Cartridge, 40MM, Practice, M385A1 making this item military unique.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Requirements for submission of first article sample.
c. Requirement for submission of design verification sample, if applicable.
d. Requirement for submission of inspection equipment designs.
e. Applicable national stock number.
f. Packaging requirements, if other than specified in Section 5.
g. Serialization requirements, if applicable.
h. Certificate of conformance for each lot or shipment of product, if applicable.
i. Government-Furnished Material or equipment, if applicable.
j. Provisions for critical characteristics controls.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. Inspection equipment, inspection method and the frequency of inspection should be submitted for approval to ARDEC, Attn: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.4 Contractor acceptance inspection equipment (AIE). Provision concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.5 Ammunition lot numbers. Ammunition requires ammunition data cards in accordance with MIL-STD-1168.

6.6 Definitions.

6.6.1 Critical defect. A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

6.6.2 Major defect. A major defect is a defect, other than critical, that is likely to result in failure or to reduce materially the usability of the unit of product for its intended purpose.

6.6.3 Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose or is a departure from established standards having little bearing on the effective use or operation of the unit.

6.7 Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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6.9 Subject term (key word) listing.

40MM Ammunition
M16A2 Metallic Link
M169 Cartridge Case
MK19 Mod 3 Grenade Machine Gun

Custodian:
Army-AR
Air Force-99
Navy-OS

Preparing activity:
Army-AR
(Project 1310-2006-018)

Review activities:
Air Force-70
Navy-MC,NP
GSA-FSS

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